

Truth Table

Number of Variables:

5 Variables

Clear All

Fill All with 1

#	A	B	C	D	E	Output
0	0	0	0	0	0	0 1 X
1	0	0	0	0	1	0 1 X
2	0	0	0	1	0	0 1 X
3	0	0	0	1	1	0 1 X
4	0	0	1	0	0	0 1 X
5	0	0	1	0	1	0 1 X
6	0	0	1	1	0	0 1 X
7	0	0	1	1	1	0 1 X
8	0	1	0	0	0	0 1 X
9	0	1	0	0	1	0 1 X
10	0	1	0	1	0	0 1 X
11	0	1	0	1	1	0 1 X
12	0	1	1	0	0	0 1 X
13	0	1	1	0	1	0 1 X
14	0	1	1	1	0	0 1 X
15	0	1	1	1	1	0 1 X
16	1	0	0	0	0	0 1 X
17	1	0	0	0	1	0 1 X
18	1	0	0	1	0	0 1 X
19	1	0	0	1	1	0 1 X
20	1	0	1	0	0	0 1 X
21	1	0	1	0	1	0 1 X
22	1	0	1	1	0	0 1 X
23	1	0	1	1	1	0 1 X
24	1	1	0	0	0	0 1 X
25	1	1	0	0	1	0 1 X
26	1	1	0	1	0	0 1 X
27	1	1	0	1	1	0 1 X
28	1	1	1	0	0	0 1 X
29	1	1	1	0	1	0 1 X
30	1	1	1	1	0	0 1 X
31	1	1	1	1	1	0 1 X

Click buttons to set output values:

0

or

1

or

X

X represents "don't care" condition (can be either 0 or 1)

Simplified K-Map Result

SOP Form

POS Form

A=0

DEBC

0000

011

110

101

000

014

1112

108

000

015

1113

109

003

017

1115

1011

002

016

1114

1010

A=1

DEBC

0016

0117

1119

1018

0020

0121

1123

1022

0028

0129

1131

1030

0024

0125

1127

1026

Simplified Boolean Expression

AE + BD + CD'E'

Original Groups

Original

AB'C'D'E + AB'C'DE + AB'CD'E + AB'CDE + ABC'D'E + ABC'DE + ABCD'E + ABCDE

Minterms

(17, 19, 21, 23, 25, 27, 29, 31)

Simplified

AE

Original

A'BC'DE' + A'BC'DE + A'BCDE' + A'BCDE + ABC'DE' + ABC'DE + ABCDE' + ABCDE

Minterms

(10, 11, 14, 15, 26, 27, 30, 31)

Simplified

BD

Original

A'B'CD'E' + A'BCD'E' + AB'CD'E' + ABCD'E'

Minterms

(4, 12, 20, 28)

Simplified

CD'E'

Logic Circuit

A

E

B

D

C

D

E

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